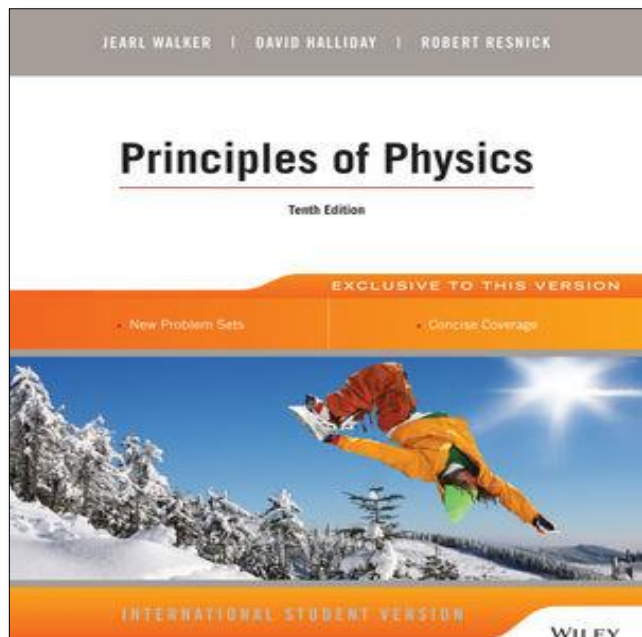


General Physics II

第三教学楼106



Halliday & Resnick

**Principles of Physics extended
International Student Version
(10th edition)**

Jearl Walker, David Halliday, Robert Resnick
Wiley

QQ Group: 422152137

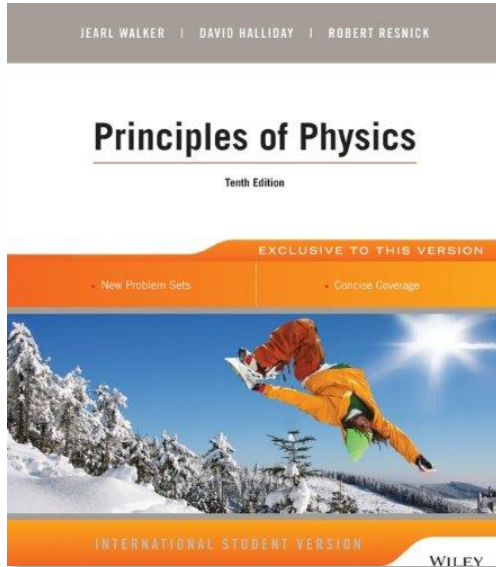
朱金龙 (Zhu Jinlong)

理学院大楼P4112

zhujl@sustech.edu.cn

Research Introduction: <http://faculty.sustech.edu.cn/zhujl/>

Textbook



Principles of Physics

International Student Version
Halliday, Resnick, Walker
10th Edition

21. Coulomb's Law
22. Electric Fields
23. Gauss' Law
24. Electric Potential
25. Capacitance
26. Current and Resistance
27. Circuits
28. Magnetic Fields
29. Magnetic Fields Due to Currents
30. Induction and Inductance
31. Electromagnetic Oscillations and Alternating Current
32. Maxwell's Equations;
~~Magnetism of Matter~~
33. Electromagnetic Waves
- ~~34. Images~~
35. Interference
36. Diffraction
37. Relativity
38. Photons and Matter Waves

Teaching outline and Grading policy

Chapter	Sections
21	1,2,3
22	1,2,3,4,5,6,7
23	1,2,3,4,5,6
24	1,2,3,4,5,6,7,8
25	1,2,3,4,5,6
26	1,2,3,4,5
27	1,2,3,4
28	1,2,3,4,5,6,7,8
29	1,2,3,4,5
30	1,2,3,4,5,6,7,8,9
31	1,2,3,4,5,6
32	1,2,3,4,5,6,7,8
33	1,2,3,4,5,6,7
34	1,2,3,4,5,6
35	1,2,3,4,5
36	1,2,3,4,5,6,7
37	1,2,3,4,5,6
38	1,2,3,4,5,6,7,8,9

**Mid-term
Exam**

**Final
Exam**

Grading Policy:

- ❖ Selected Homework: 10%,
- ❖ In class test: 10%,
- ❖ Midterm exam: 40%,
- ❖ Final Exam: 40% .

Exam arrangement:

- ❖ Midterm, chapters 21-29
April 20: 9:00~11:00
 Make up: **April 26: 19:00~21:00**
- ❖ Final exam, chapters 21-33&35-38

The sections marked in red are **not** included in examinations.

Stay Informed

- QQ group for this class: 422152137
- **All course-related material (lecture notes PDF, homework, notice, etc.) will be distributed through the above QQ group!**

Always stay tuned!

Lectures and TA information

	Mon.	Tue.	Wed.	Thu.	Fri.
上午					
					10:20-12:10 (3 4节)
下午					
		16:20-18:10 (7 8节)			

section	Mon	Tue	Wed	Thur	Fri
10:00-11:30					
14:00-16:00					

李克华

Tel: 182 7391 1983

Office: 理学院P5127

Email: likh@mail.sustech.edu.cn

Homework notice:

1. Use **blackboard** to submit, grade and return your homework.
2. Do your homework with Principles of Physics, 10th Ed [David Halliday] , download the soft copy from QQ group or from blackboard
3. Write your homework on papers and Write your NAME, student number and chapter-number(Example: Ch21-10) on the top of the paper.
4. Take photos of your homework and save them as a single pdf file and upload it to blackboard.
5. Homework submitted within 48 hours after the deadline will be graded, but with a maximum score of 50%; homework submitted 48 hours after the deadline will not be graded!
6. Check your homework from blackboard and QQ group weekly.

Homework

Chapter	24年春季大物（下）
Ch21	5, 10, 24
Ch22	6, 18, 34, 47, 54, 61
Ch23	2, 4, 9, 24, 33, 47, 49
Ch24	4, 9, 10, 24, 34, 48, 53, 67
Ch25	2, 7, 17, 21, 39, 49, 52
Ch26	4, 5, 27, 30, 46
Ch27	3, 9, 17, 37, 47, 69
Ch28	1, 17, 18, 30, 32, 46, 47
Ch29	6, 7, 16, 19, 21, 24, 26, 27
Ch30	3, 8, 13, 14, 24, 40, 43, 69
Ch31	3, 17, 23, 26, 30, 43, 58
Ch32	3, 45, 49
Ch33	8, 35, 39, 47, 59, 67
Ch35	28, 41, 42, 47, 56, 81
Ch36	17, 22, 27, 29, 37, 39
Ch37	3, 10, 11, 22, 23, 30, 43, 59
Ch38	2, 9, 17, 25, 64

SUSTech Blackboard

Chapter 21 Coulomb's Law
Chapter 22 Electric Fields
Chapter 23 Gauss's Law
Chapter 24 Electric Potential
Chapter 25 Capacitance
Chapter 26 Current and Resistance
Chapter 27 Circuits
Chapter 28 Magnetic Fields
Chapter 29 Magnetic Fields Due to Currents
Chapter 30 Induction and Inductance
Chapter 31 Electromagnetic Oscillations and Alternating Current
Chapter 32 Maxwell's Equations; ~~Magnetism of Matter~~
Chapter 33 Electromagnetic Waves
~~Chapter 34 Images~~
Chapter 35 Interference
Chapter 36 Diffraction
Chapter 37 Relativity
Chapter 38 Photons and Matter Waves
~~Chapter 39 More About Matter Waves~~

我的作业

课堂测验

我的成绩

>>查看/完成测验

Sample Calendar

教学日历仅供参考，各位老师可根据授课实际情况调整课时安排														
Month	Week	Mon	Tue	Wed	Thur	Fri	Sat	Sun	补课安排	Lecture1	Lecture2	Tutor	备注	
Feb.	1	19	20	21	22	23	24	25	退选课周	课前介绍 +Ch21~22 2H	Ch22 1H	Ch22 1H		
	2	26	27	28	29	1	2	3		Ch23 2H	Ch23 1H	Ch23 1H		
Mar.	3	4	5	6	7	8	9	10			Ch24 2H	Ch24 1H	Ch24 1H	
	4	11	12	13	14	15	16	17		Ch25 2H	Ch25 1H	Ch25 1H		
	5	18	19	20	21	22	23	24		Ch26 2H	Ch27 2H			
	6	25	26	27	28	29	30	31		Ch27&28 2H	Ch 28 1H	Ch27 1H		
Apr.	7	1	2	3	4	5	6	7	4月7日上单周 周五的课	Ch28&29 2H	Ch29 1H	Ch28 1H		
	8	8	9	10	11	12	13	14		Ch29&30 2H	Ch30 1H	Ch29 1H	期中考试： 4月20号 9:00-11:00 Ch 21~29	
	9	15	16	17	18	19	20	21	Ch30&31 2H	Ch31 1H	Ch30 1H	补考时间4月26号晚19:00-21:00		
	10	22	23	24	25	26	27	28	4月28日上单周 周四的课	Ch31&32 2H			Ch31&32试卷 2H	
May.	11	29	30	1	2	3	4	5	5月11日上单周 周三的课	Ch33 2H	Ch33 1H	Ch33 1H		
	12	6	7	8	9	10	11	12			Ch35 2H			
	13	13	14	15	16	17	18	19		Ch35&36 2H	Ch36 1H	Ch35 1H		
	14	20	21	22	23	24	25	26		Ch36&37 2H	Ch37 1H	Ch36 1H		
	15	27	28	29	30	31	1	2		Ch37&38 2H	Ch38 1H	Ch37 1H		
June.	16	3	4	5	6	7	8	9		Ch38&39 2H		答疑	39章有时间可选择性讲，但不计入考试内容	
	17	10	11	12	13	14	15	16	期末考试周	期末考试周				
	18	17	18	19	20	21	22	23						
说明：	周四、周五上课的班级会少一次课，共62课时，其他班级64课时													

Demo List

21	Electric friction	经典静电材料套装
25	electrostatic experiment	静电演示组合实验仪器
25	Capacitance	可变电容实验包
28	Magnetic force on a current-carrying wire	磁铁对通电直导线作用力的演示
30	eddy current	磁阻尼摆演示仪、楞次定律演示仪及延长管
30	lenz' law	跳环型楞次定律演示仪
30	Induction	通电线圈间相互作用力演示仪
33	polarization	手持大偏振片
35-36	Interference and Diffraction of Light	单双缝实验包

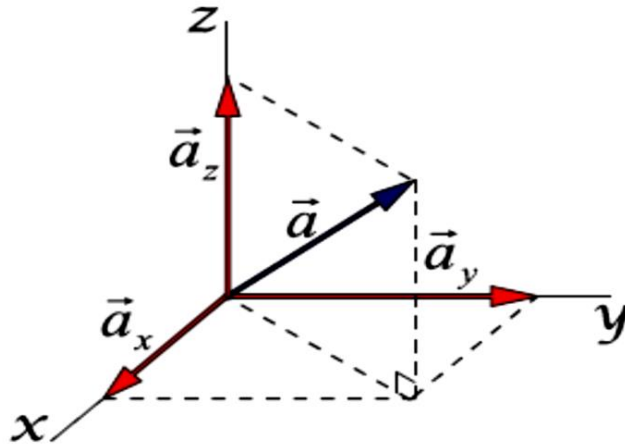
To Succeed

- Come to class, and bring your mind
- Read textbook before class
- Take notes in class
- Study lecture notes and textbook after class
- Do homework and ask help in office hours
- **6-10 hours per week** in addition to class

Recap: Components of Vectors

Usually, vectors are expressed according to coordinate system. Each vector can be expressed in terms of *components*.

The most common coordinate system: **Cartesian**



$$\vec{a} = \vec{a}_x + \vec{a}_y + \vec{a}_z$$

Magnitude of $\vec{a} = |\vec{a}| = a$,

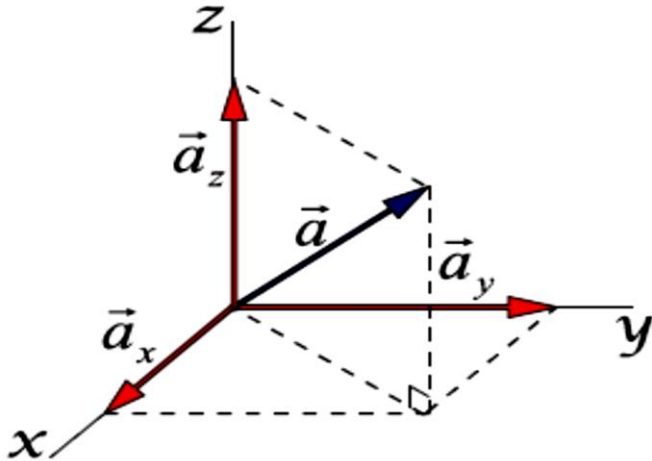
$$a = \sqrt{a_x^2 + a_y^2 + a_z^2}$$

Recap: Components of Vectors

Unit vectors have magnitude of 1

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|} = \text{unit vector along } \vec{a} \text{ direction}$$

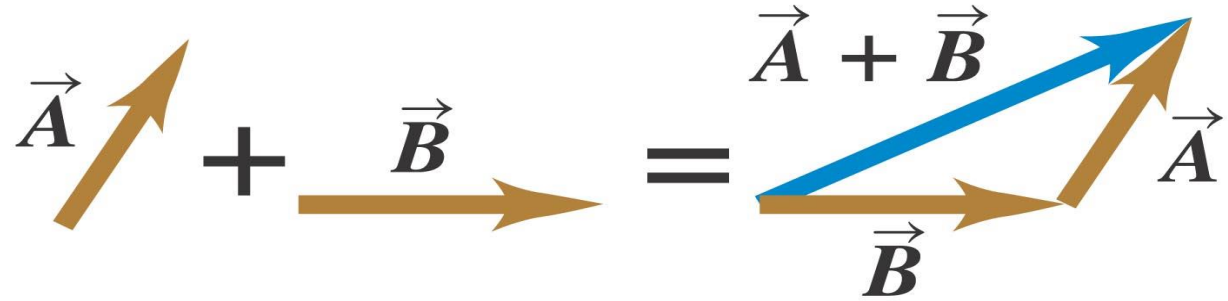
$\hat{i}$ $\hat{j}$ $\hat{k}$ are unit vectors along
 $\updownarrow$ $\updownarrow$ $\updownarrow$
 x y z directions



$$\vec{a} = a_x \hat{i} + a_y \hat{j} + a_z \hat{k}$$

Recap: Vector Operation

Addition



Scalar Product (dot product)

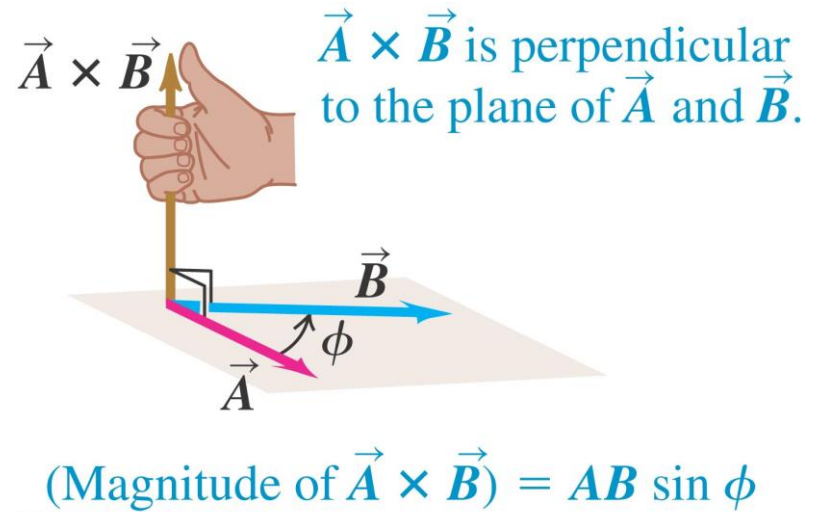
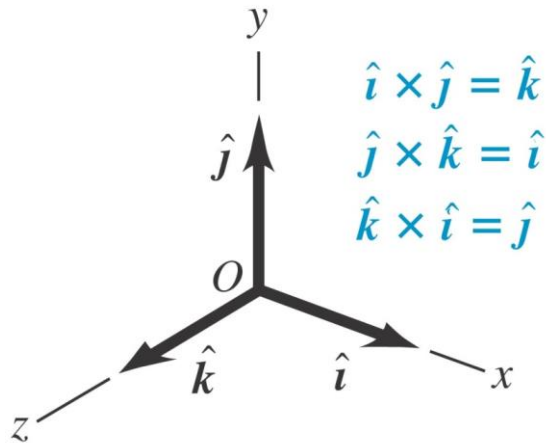
$$\text{Scalar product } \vec{A} \cdot \vec{B} = AB \cos \phi$$



Recap: Vector Operation

Vector Product (cross product)

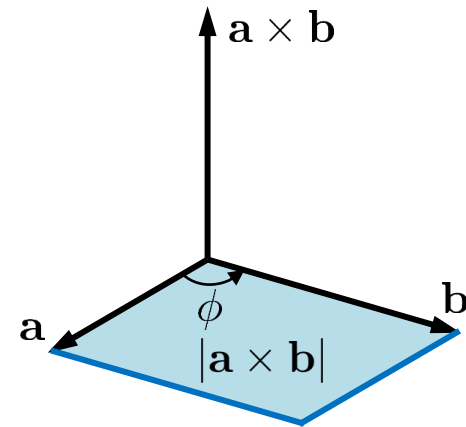
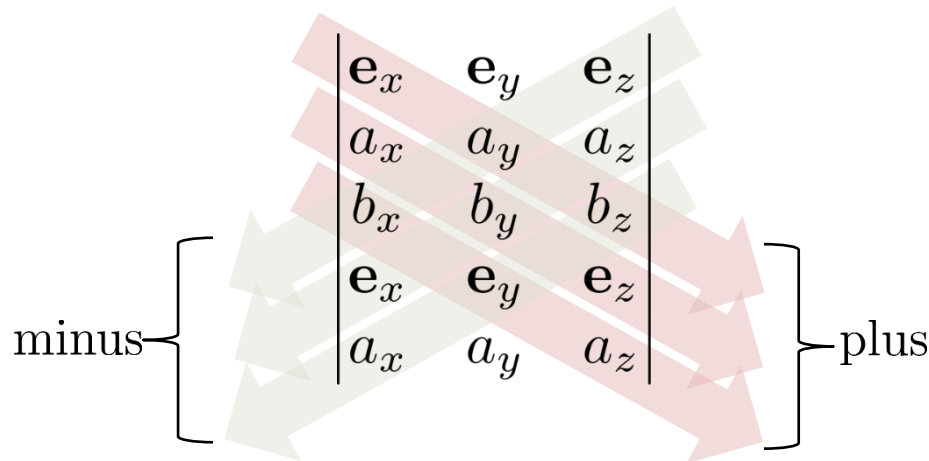
A right-handed coordinate system



Components of the Cross Product

To get the x, y, z components of $\mathbf{c}$, according to the definition of a cross product, we can use the following determinant form:

$$\begin{aligned}\mathbf{c} = \mathbf{a} \times \mathbf{b} &= \begin{vmatrix} \mathbf{e}_x & \mathbf{e}_y & \mathbf{e}_z \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix} \\ &= (a_y b_z \mathbf{e}_x + a_z b_x \mathbf{e}_y + a_x b_y \mathbf{e}_z) - (a_z b_y \mathbf{e}_x + a_x b_z \mathbf{e}_y + a_y b_x \mathbf{e}_z) \\ &= (a_y b_z - a_z b_y) \mathbf{e}_x + (a_z b_x - a_x b_z) \mathbf{e}_y + (a_x b_y - a_y b_x) \mathbf{e}_z\end{aligned}$$



Recap: Vector Operation

Vector differential operator

Delta or Nabla

$$\nabla \equiv \frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k}$$

$$\nabla u = \left(\frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k} \right) u = \frac{\partial u}{\partial x} \vec{i} + \frac{\partial u}{\partial y} \vec{j} + \frac{\partial u}{\partial z} \vec{k}$$

$$\nabla \cdot \vec{A} = \left(\frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k} \right) (A_x \vec{i} + A_y \vec{j} + A_z \vec{k})$$

$$= \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$\nabla \times \vec{A} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

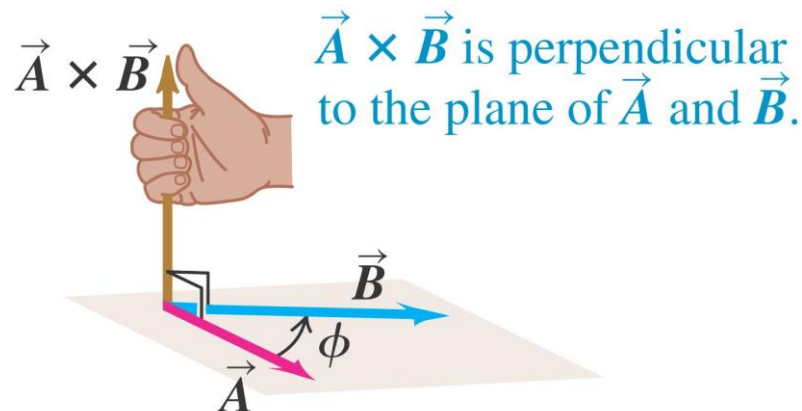
$$= \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \vec{i} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \vec{j} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \vec{k}$$

因此，数量场的梯度与矢量场的散度及旋度可用 ∇ 算子表示

$$\text{grad} u = \nabla u$$

$$\text{div} \vec{A} = \nabla \cdot \vec{A}$$

$$\text{rot} \vec{A} = \nabla \times \vec{A}$$



$$(\text{Magnitude of } \vec{A} \times \vec{B}) = AB \sin \phi$$

Recap: Newton's Laws of Motion

Newton's laws of motion

Newton's First Law: If no force acts on a body, then the body's velocity cannot change

Newton's Second Law: $\vec{F} = m\vec{a}$

Newton's laws of motion allow us to analyze many kinds of motion.

Recap: Conservation Laws

Conservation of energy

The total energy of an isolated system cannot change

In an isolated system where only conservative forces cause energy changes, the kinetic energy and potential energy can change, but their sum cannot change

$$K_i + U_i = K_f + U_f$$

Recap: Conservation Laws

Conservation of momentum

Conservation of linear momentum

If no net external force acts on a system of particles, the total linear momentum of the system cannot change

$$\vec{P} = m_1 \vec{v}_1 + m_2 \vec{v}_2 + \dots + m_n \vec{v}_n = \text{const}$$

Conservation of angular momentum

If the net external torque acting on a system is zero, the angular momentum of the system remains constant, no matter what changes take place within the system

$$\vec{L}_i = \vec{L}_f$$

Recap: Gravitational Force

<https://v.qq.com/x/page/I0735ckbosq.html>

https://www.iqiyi.com/w_19s81052b9.html

Different forces:

gravitational force

friction

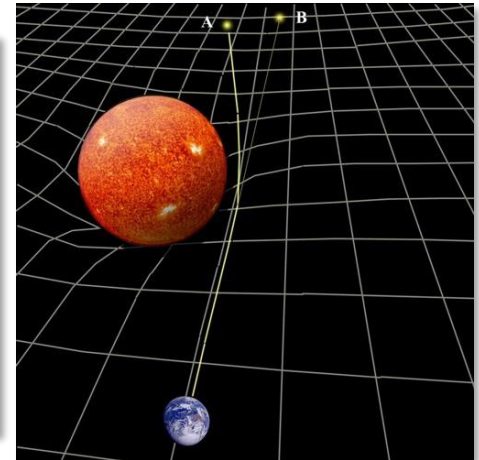
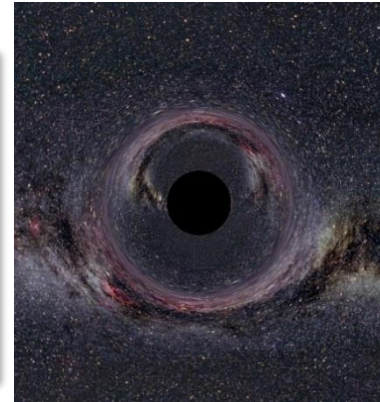
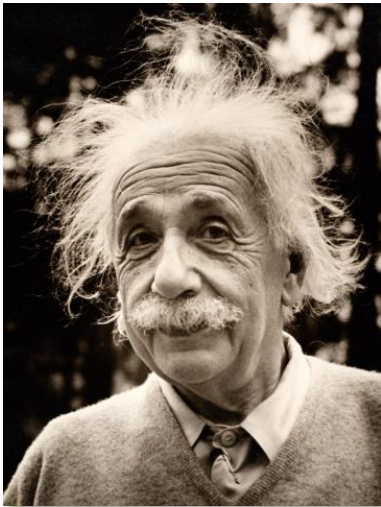
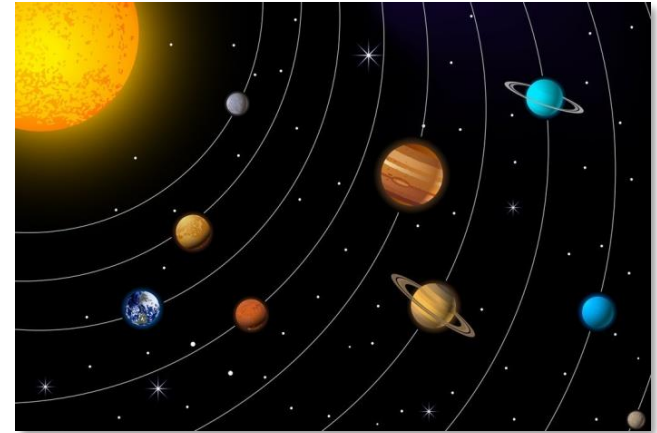
normal force

tension

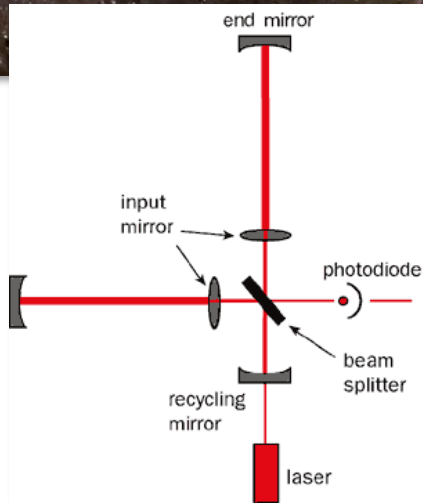
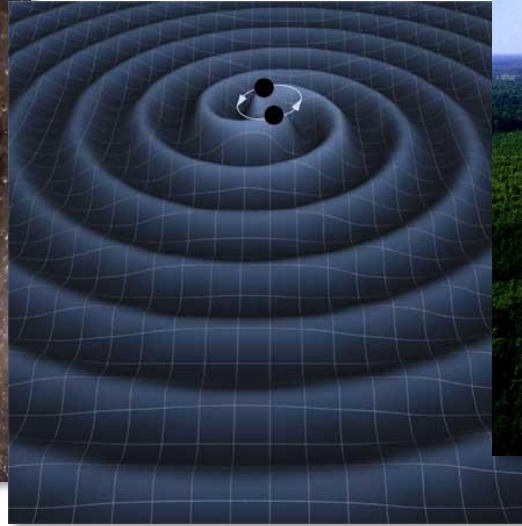
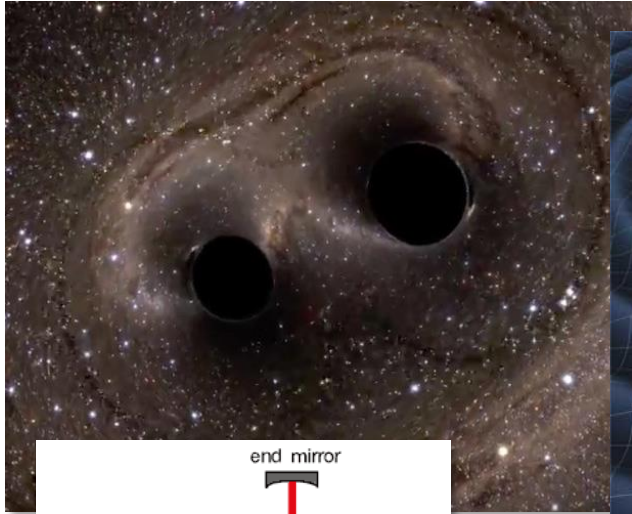
spring force

buoyancy

...



Gravitational Wave and LIGO Experiment



Official announcement: <http://www.youtube.com/watch?v=aEPlwEJmZyE>

Brian Greene explains GW: http://www.youtube.com/watch?v=s06_jRK939I

New Yorker article:

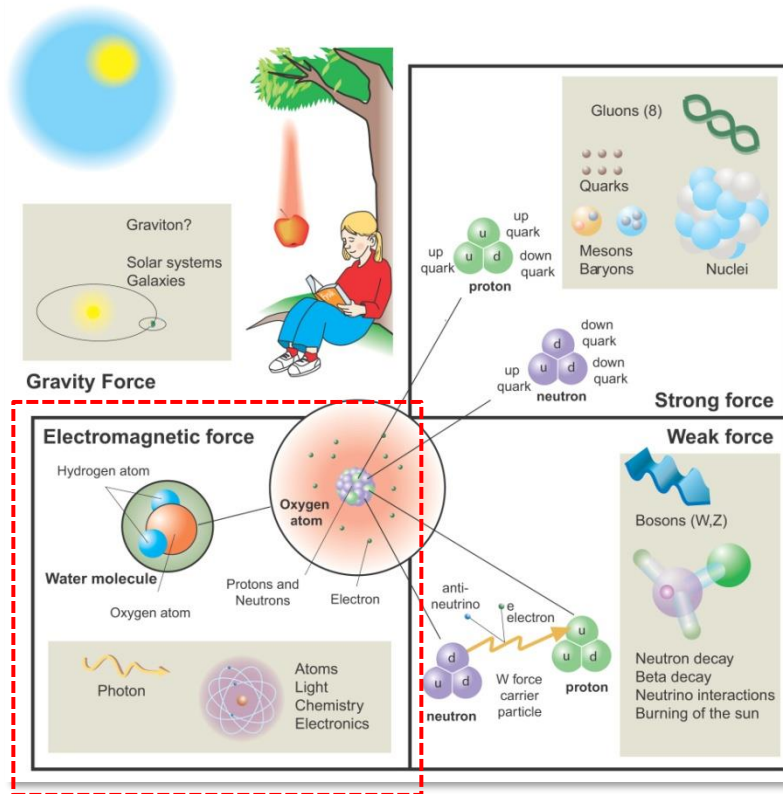
<http://www.newyorker.com/tech/elements/gravitational-waves-exist-heres-how-scientists-finally-found-them>

Chinese version: <https://zhuanlan.zhihu.com/p/467631258>

<https://v.qq.com/x/page/x0561ap1nlf.html>

Four Fundamental Forces

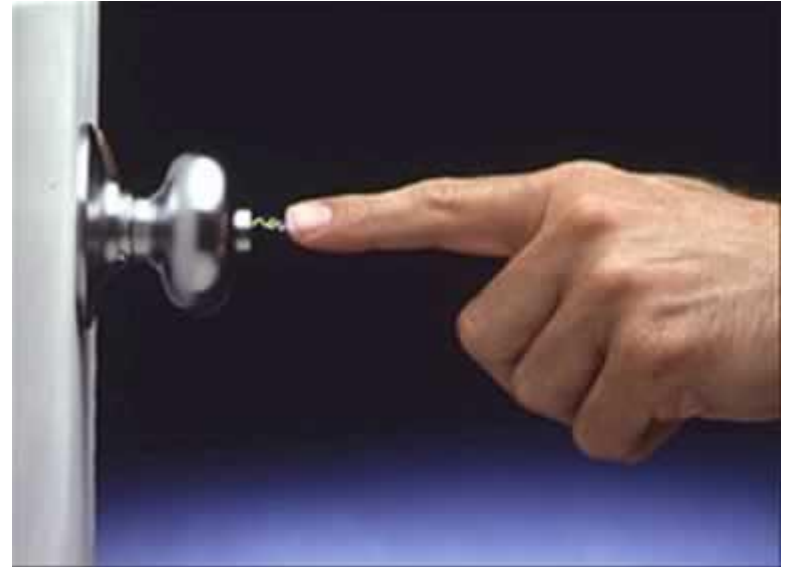
Different forces: friction, normal force, tension, spring force, buoyancy, gravitational force...



Fundamental Forces				
Strong	Force which holds nucleus together	Strength 1	Range (m) 10^{-15} (diameter of a medium sized nucleus)	Particle gluons, π (nucleons)
Electro-magnetic		Strength $\frac{1}{137}$	Range (m) Infinite	Particle photon mass = 0 spin = 1
Weak	neutrino interaction induces beta decay	Strength 10^{-6}	Range (m) 10^{-18} (0.1% of the diameter of a proton)	Particle Intermediate vector bosons W^+ , W^- , Z^0 , mass > 80 GeV spin = 1
Gravity		Strength 6×10^{-39}	Range (m) Infinite	Particle graviton ? mass = 0 spin = 2

Coulomb's Law

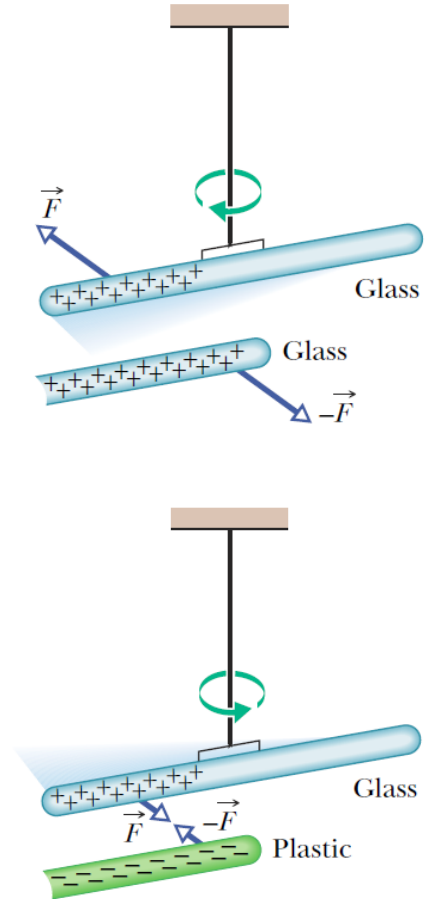
Electric Phenomena in Nature



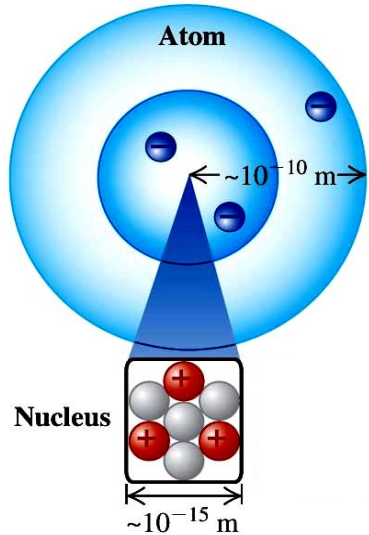
Positive and Negative Electric Charges



- * There are two types of electric charges, positive (+) and negative (-).
- * Objects bearing electric charges of the same sign *repel* each other; objects bearing electric charges of the opposite signs attract each other.



Electric Charge and Structure of Matter



 **Proton:** Positive charge

 **Neutron:** No charge

 **Electron:** Negative charge

The charges of the electron and proton are equal in magnitude.

Electric Charge is an intrinsic characteristic of the fundamental particles making up objects around us (including us).

The ordinary matter consists of three (only!) particles:

	electron (e)	proton (p)	neutron (n)
mass	$9.11 \times 10^{-31} \text{ kg}$	$1.67 \times 10^{-27} \text{ kg}$	$1.67 \times 10^{-27} \text{ kg}$
charge	$-1.60 \times 10^{-19} \text{ C}$	$1.60 \times 10^{-19} \text{ C}$	0.00

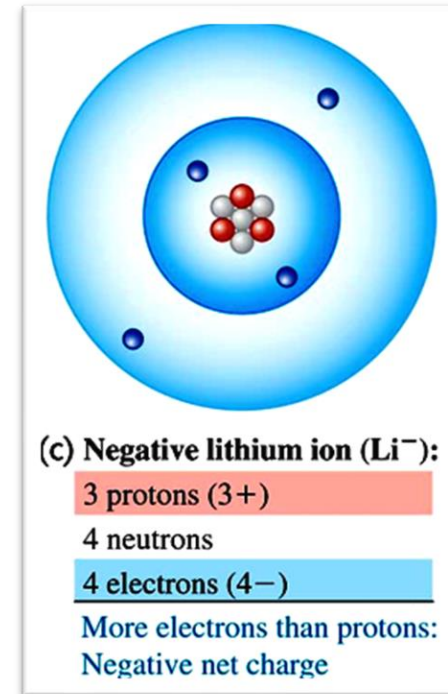
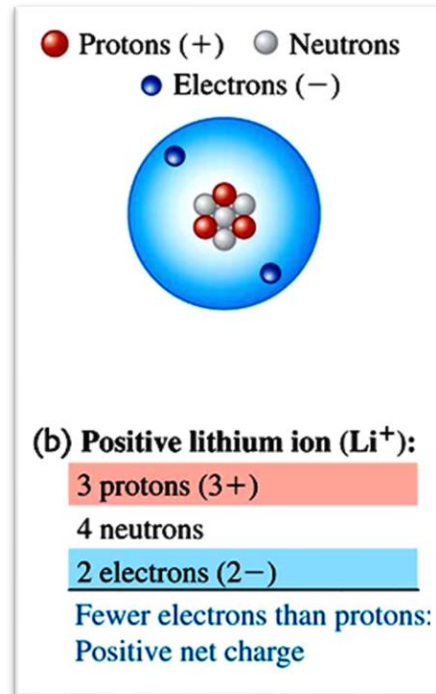
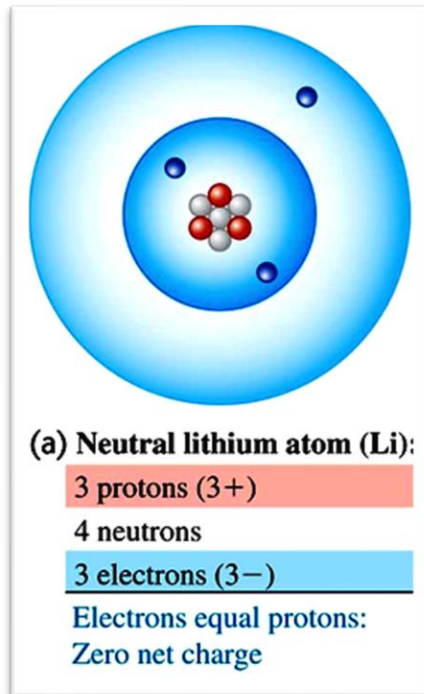
The SI unit of electric charge is Coulomb.

核裂变与核聚变有什么区别？原理是什么？
铀在其中扮演怎样的角色？ - 知乎 (zhihu.com)

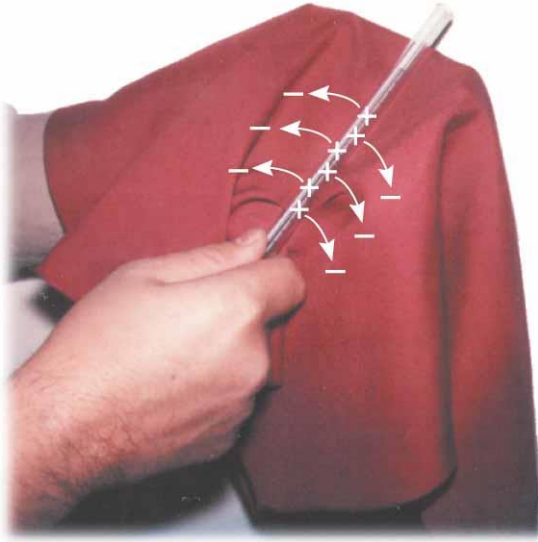
<https://www.zhihu.com/question/362231802>

Atoms and Ions

- A neutral atom has the same number of protons as electrons.
- A **positive ion** is an atom with one or more electrons removed.
- A **negative ion** has gained one or more electrons.



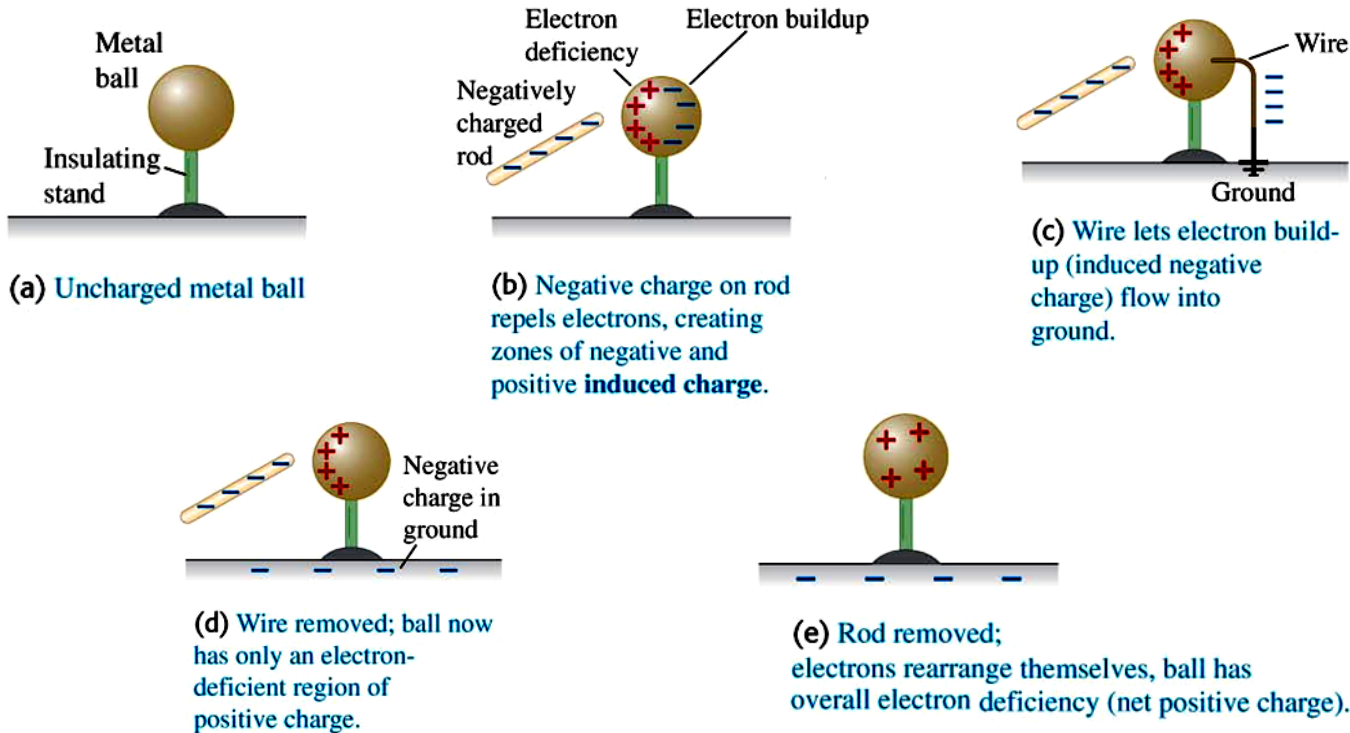
Conservation of Charge



- * Total charge in an **isolated** system is conserved
i.e. charge is not created but transferred from one object to another (for example, from a glass rod to a piece of silk when rubbing one against the other).

Charging by induction

- The negative rod is able to charge the metal ball without losing any of its own charge. This process is called **charging by induction (感应)**.

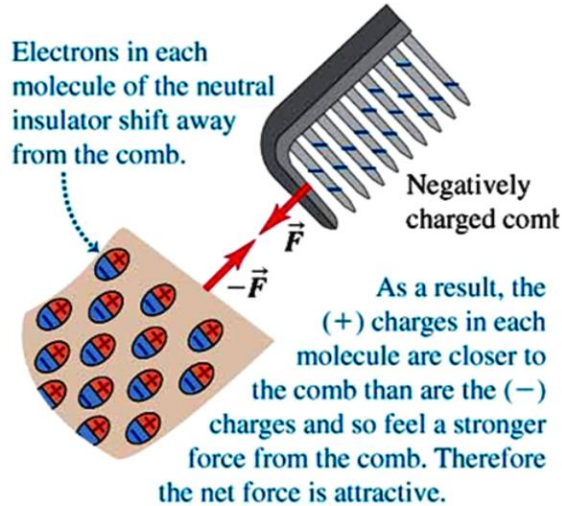


Why Do Uncharged Objects Feel Electric Force?

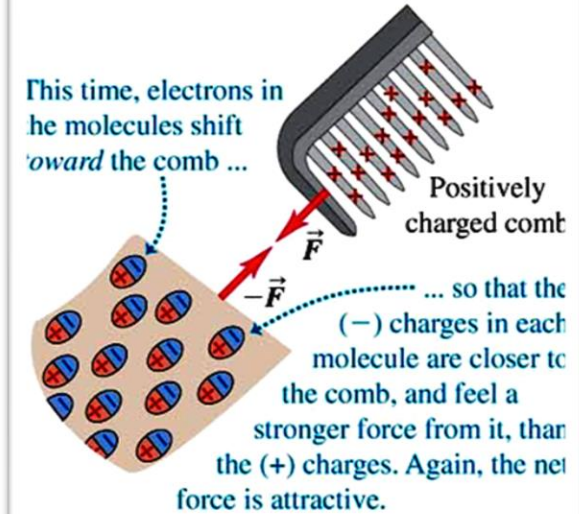
(a) A charged comb picking up uncharged pieces of plastic



(b) How a negatively charged comb attracts an insulator



(c) How a positively charged comb attracts an insulator



- The charge within an insulator can shift slightly. As a result, two neutral objects can exert electric forces on each other.

Quantization of Charge

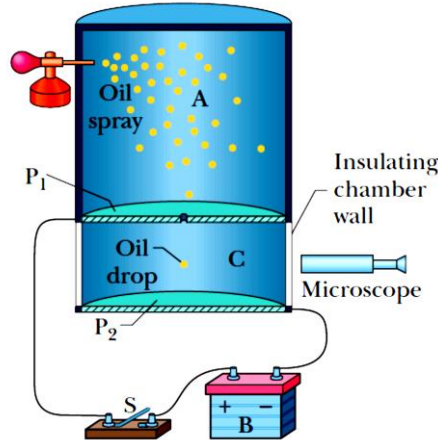
Experiments show that any positive or negative charge q that can be detected can be written as

$$q = n \cdot e, \quad n = \pm 1, \pm 2, \pm 3, \dots$$
$$e = 1.60 \times 10^{-19} \text{ C}$$

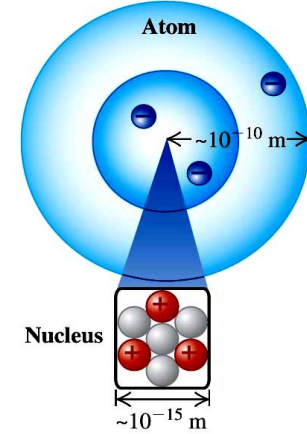
where e is the **elementary charge**.



Robert A. Millikan
1923 Nobel Prize



R. A. Millikan, *Phys. Rev.* 2, 109 (1913)



As of 2014, the accepted value for the elementary charge is

$$e = 1.602176565(35) \times 10^{-19} \text{ C}$$

This is the charge of one electron.

Millikan Oil Drop Experiment



MILLIKAN'S OIL DROP EXPERIMENT

MILLIKAN'S OIL DROP EXPERIMENT

Millikan's Oil Drop Experiment

With the field off, drop reaches a terminal velocity

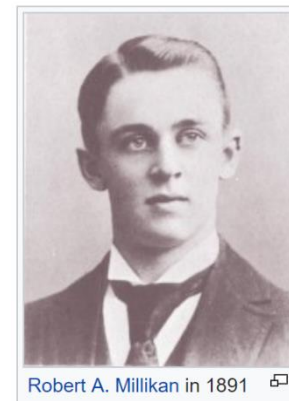
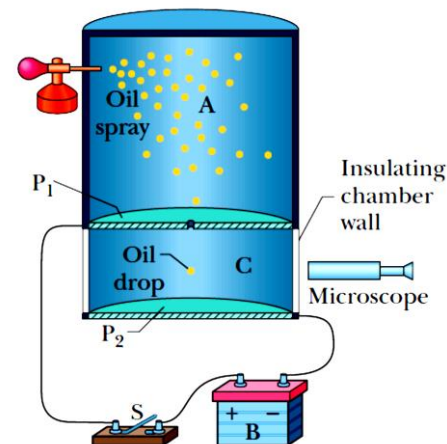
drag force: $F_d = 6\pi r\eta v_1$

weight: $w = \frac{4\pi}{3} r^3 (\rho - \rho_{air}) g$

$$F_d = w \quad \rightarrow \quad r^2 = \frac{9\eta v_1}{2(\rho - \rho_{air})g}$$

With the field on, drop reaches another terminal velocity

$$qE - w = 6\pi r\eta v_2$$



Coulomb's Law



Charles de Coulomb



Henry Cavendish

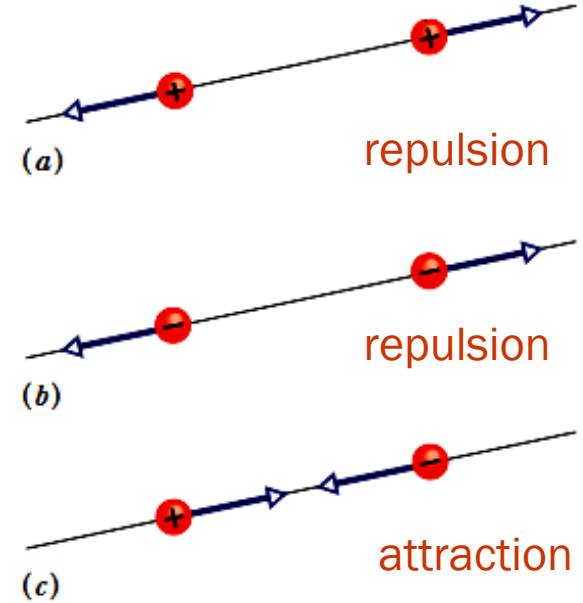
The electrostatic force between two charges q_1 and q_2 separated by a distance r has the magnitude:

$$F = k \frac{|q_1 q_2|}{r^2}$$

$$k = \frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2$$

k is the electromagnetic constant
 ϵ_0 is the vacuum permittivity constant



$$F = k \frac{|q_1||q_2|}{r^2} \quad \Rightarrow \quad [F] = [k] \frac{[q_1][q_2]}{[r]^2}$$

$$[Newton] = [k] \frac{[Coulomb][Coulomb]}{[meter]^2}$$

Then the Coulomb constant unit is

$$[k] = N \cdot m^2 / C^2$$

Comparing Gravitational and Electrostatic Forces

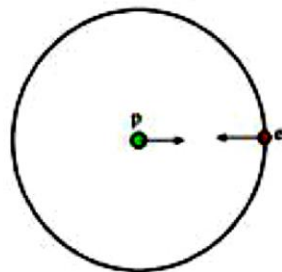
Let's calculate the ratio of the electrical to the gravitational force inside a hydrogen atoms.

Newton's law of gravitation:

$$F_g = \frac{Gm_1m_2}{r^2}$$

Coulomb's law:

$$F_e = \frac{kq_1q_2}{r^2}$$



$$F_g = \frac{GM_pM_e}{R^2}, F_e = \frac{k_e q_p q_e}{r^2}$$

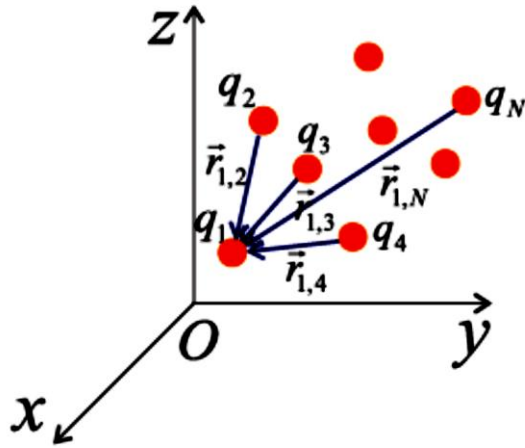
$$\frac{F_e}{F_g} = \frac{\frac{k_e q_p q_e}{\cancel{r^2}}}{\frac{GM_p M_e}{\cancel{r^2}}} = \frac{k_e q_p q_e}{GM_p M_e}$$

$$\frac{F_e}{F_g} = \frac{(9 \times 10^9)(1.6 \times 10^{-19})^2}{(6.67 \times 10^{-11})(1.67 \times 10^{-27})(9.11 \times 10^{-31})}$$
$$= 2.27 \times 10^{39}$$

So we can forget gravity as compared to electrostatic forces - at least on the atomic scale

- * The gravitational force essentially plays NO role in atomic and molecular systems.
- * Macroscopic objects are neutral or nearly neutral (net electric charge is close to zero). Therefore the gravitational force dominates for macroscopic objects with huge masses.

Principle of Superposition: Multi-charge System



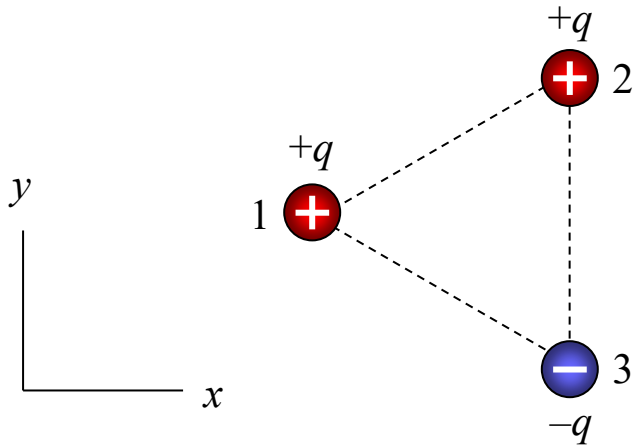
Principle of Superposition

The net electrostatic force experienced by charge q_1 is the **vector sum** of all forces exerted on q_1 by other charges:

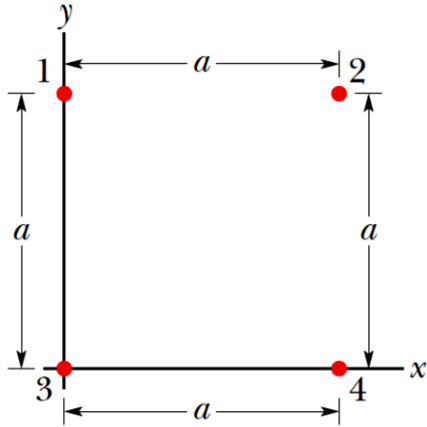
$$\vec{F}_1 = \vec{F}_{1,2} + \vec{F}_{1,3} + \dots + \vec{F}_{1,N}$$

What's the direction of the net electrostatic force felt by charge 1?

- A. +x direction.
- B. -x direction.
- C. +y direction.
- D. -y direction. ✓
- E. none of the above



Example Problem



Four particles form a square. The charges are

$q_1 = q_4 = Q$, and $q_2 = q_3 = q$.

- (a) What is Q/q if the net electrostatic force on particles 2 and 3 is zero?
- (b) Is there any value of q that makes the electrostatic force on each of the four particles zero?

Solution

(a) We calculate the net force on particle 2 (particle 3 would be the same) by:

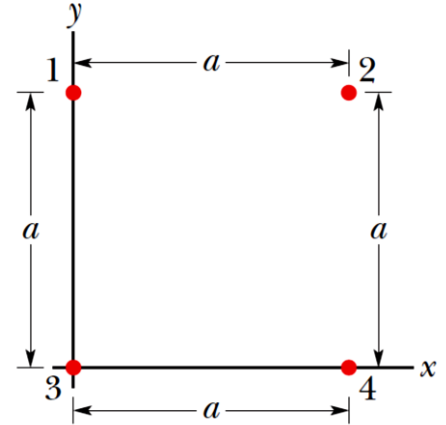
$$\begin{aligned}\mathbf{F}_2 &= \sum_i \mathbf{F}_{i2} = \mathbf{F}_{12} + \mathbf{F}_{32} + \mathbf{F}_{42} \\ &= \frac{1}{4\pi\epsilon_0} \left(\frac{Qq}{a^2} \mathbf{e}_x + \frac{q^2}{(\sqrt{2}a)^2} \mathbf{e}_{32} + \frac{Qq}{a^2} \mathbf{e}_y \right)\end{aligned}$$

Since $\mathbf{e}_{32} = \frac{\sqrt{2}}{2} \mathbf{e}_x + \frac{\sqrt{2}}{2} \mathbf{e}_y$,

$$\mathbf{F}_2 = \frac{1}{4\pi\epsilon_0} \left[\left(\frac{Qq}{a^2} + \frac{\sqrt{2}q^2}{4a^2} \right) \mathbf{e}_x + \left(\frac{Qq}{a^2} + \frac{\sqrt{2}q^2}{4a^2} \right) \mathbf{e}_y \right].$$

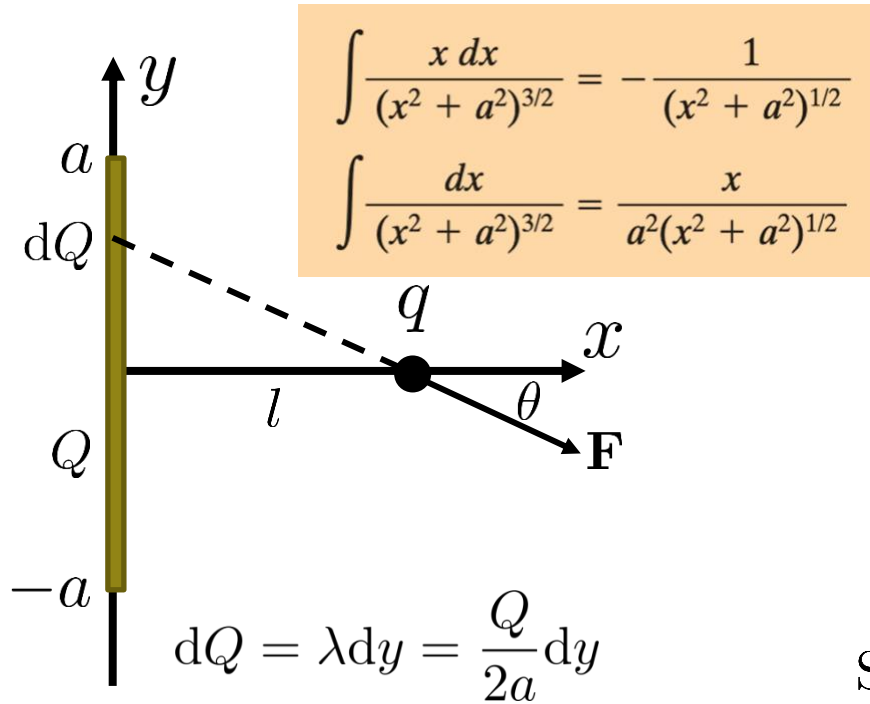
In order for $\mathbf{F}_2 = 0$, we must have

$$\frac{Qq}{a^2} + \frac{\sqrt{2}q^2}{4a^2} = 0, \quad \text{i.e., } q = 0 \quad \text{or} \quad \frac{Q}{q} = -\frac{\sqrt{2}}{4}.$$



A More Difficult Problem

Example: A rod with length $2a$ and uniformly distributed charge Q is placed on the y -axis, spanning the region $\{(x, y) | x = 0, -a < y < a\}$. Find the electric force felt by a charge q at point $(l, 0)$.



$$\begin{aligned} F_x &= \int_{-a}^a \frac{1}{4\pi\epsilon_0} \frac{q dQ}{l^2 + y^2} \cos \theta \\ &= \frac{1}{4\pi\epsilon_0} \frac{Qq}{2a} \int_{-a}^a \frac{dy}{l^2 + y^2} \frac{l}{\sqrt{l^2 + y^2}} \\ &= \frac{Qql}{8\pi\epsilon_0 a} \frac{2a}{l^2 \sqrt{a^2 + l^2}} \\ &= \frac{Qq}{4\pi\epsilon_0 l \sqrt{a^2 + l^2}}; \end{aligned}$$

$$F_y = \int_{-a}^a \frac{1}{4\pi\epsilon_0} \frac{q dQ}{l^2 + y^2} \sin \theta = 0$$

$$\text{So, } \mathbf{F} = \frac{Qq}{4\pi\epsilon_0 l \sqrt{a^2 + l^2}} \mathbf{e}_x$$